

Compilers Assignment 2

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1 Very Busy Expression

1.1 Definition

An expression is considered to be **very busy** in a specific instance p when said expression is evaluated on all path from p to the exit node without having its operands redefined along the way.

1.2 Data Flow Analysis

Backwards flow analysis proves efficient here. We need to use

$$\begin{aligned} In(B) &= f(Out(B)) \\ In(B) &= Use(B)U(Out(B) - Def(B)) \end{aligned}$$

$$\begin{aligned} Out(B) &= \wedge In(S(B)) \\ \wedge &= \cap \end{aligned}$$

Domain	Set of all expressions
Direction	Backwards $In(B) = f(Out(B))$ $Out(B) = \wedge In(Succ(B))$
Transfer function	$In(B) = Use(B)U(Out(B) - Def(B))$
Meet operation	$\cap$
Boundary Condition	$In(Exit) = \emptyset$
Initial Interior Points	$In(B) = U$

2 Dominator Analysis

2.1 Definition

We say that a node B dominates a node N if all paths from the entry to node N pass through B . By this definition we can say that all blocks trivially dominate themselves.

2.2 Data Flow Analysis

Domain	Set of all nodes of the Graph
Direction	Forward $Out(B) = f(In(B))$ $In(B) = \wedge Out(Prev(B))$
Transfer function	$Out(B) = Gen(B)U(In(B) - Kill(B))$
Meet operation	$\cap$
Boundary Condition	$Out(Entry) = \{Entry\}$
Initial Interior Points	$Out(B) = U$

3 Constant Propagation